



Applied Avionics

NEXSYS LYNK Integrated Signal Processor
Capabilities and Use Cases

Bing Jaculbe

Vice President-Technology & Product Development

*25+ years leading certified software-driven avionics
product development*

Colin McGuire

Technical Sales

*6+ years working directly with AAI customers
as a solutions provider*

Alexa Kirk

Software Development Lead

*Subject matter expert in FAA, DoD, and
Homeland Security software technologies*

Team of Applications Engineers

*Highly knowledgeable in avionics systems design
to solve your unique integration challenges*

Legacy interfaces need to coexist seamlessly with modern architectures

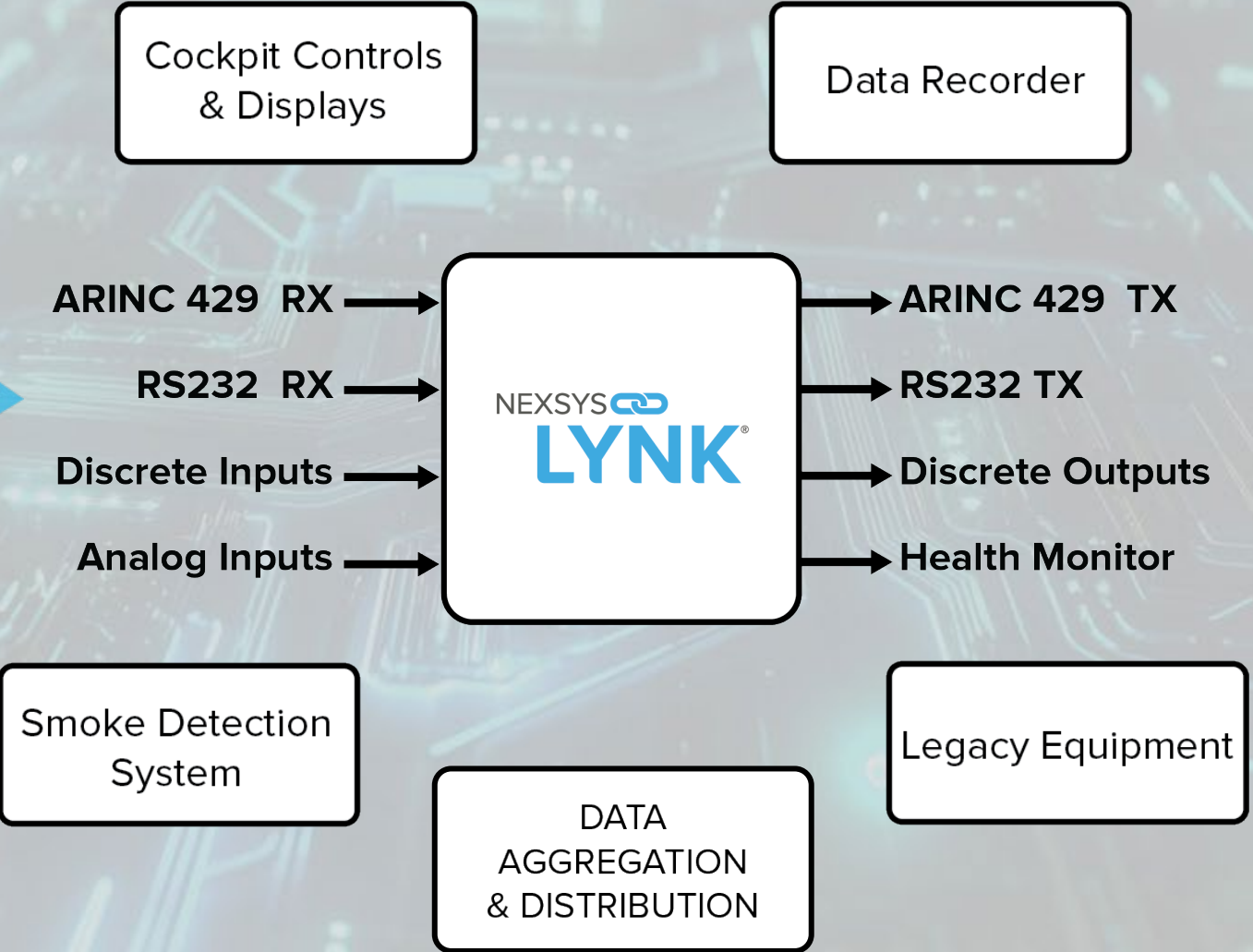
Multiple onboard devices that weren't designed to talk to each other

Signals need filtering, translating, or converting

Increased need for real-time logic/computations without adding complexity

Certification burden

Solving these challenges is exactly why LYNK exists...



Problems Solved



Dedicated Team of
Application Engineers



Solutions Meet DO-160 and
DO-178 DAL A Standards



TSO C113b Certified

GPS Message Error Detection/Correction Display

FMS Navigation Database Alert Display Unit

UTC Clock to Flight Data Recorder

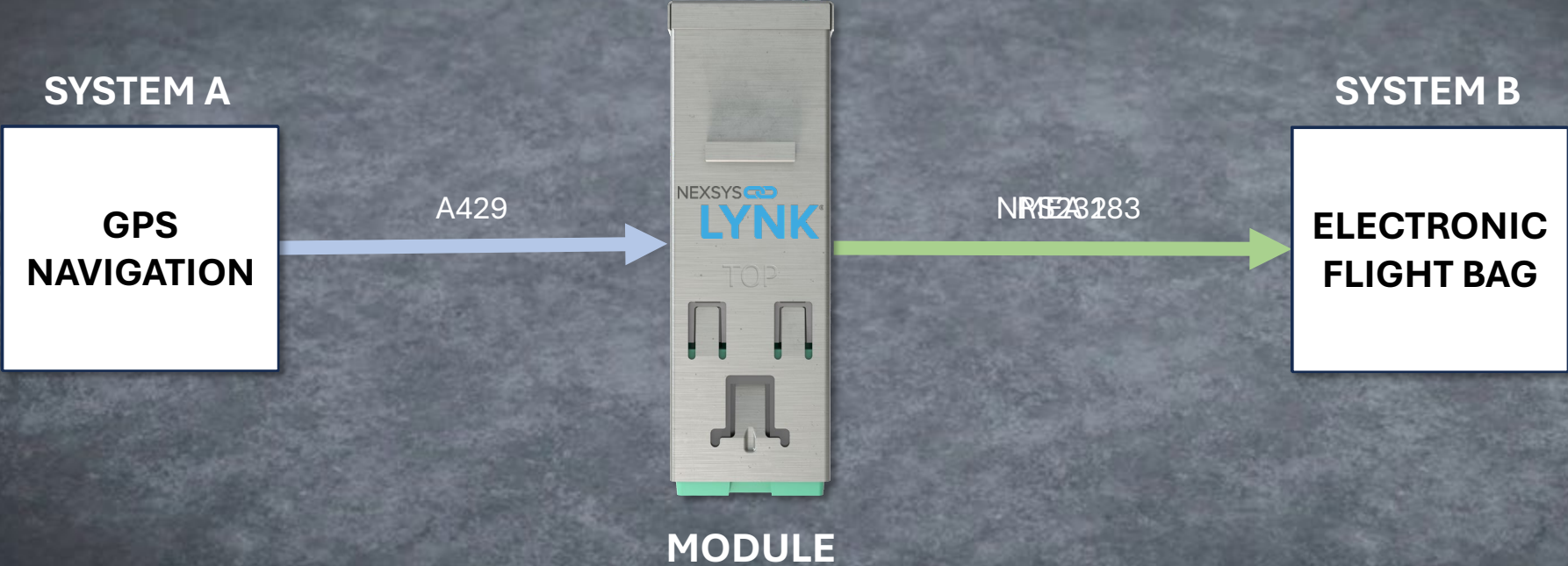
Radio Altitude Display Unit

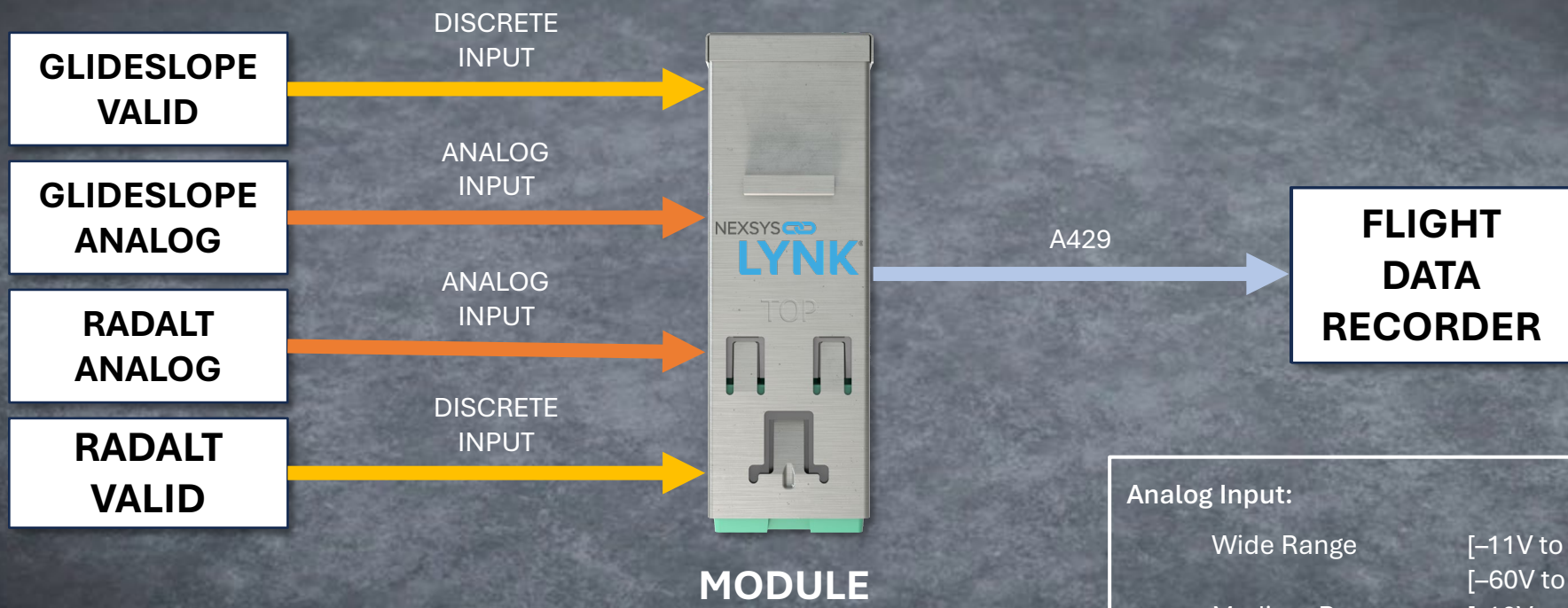
Direction Finding Smoke Detection System Computer

Door Sensor Monitoring

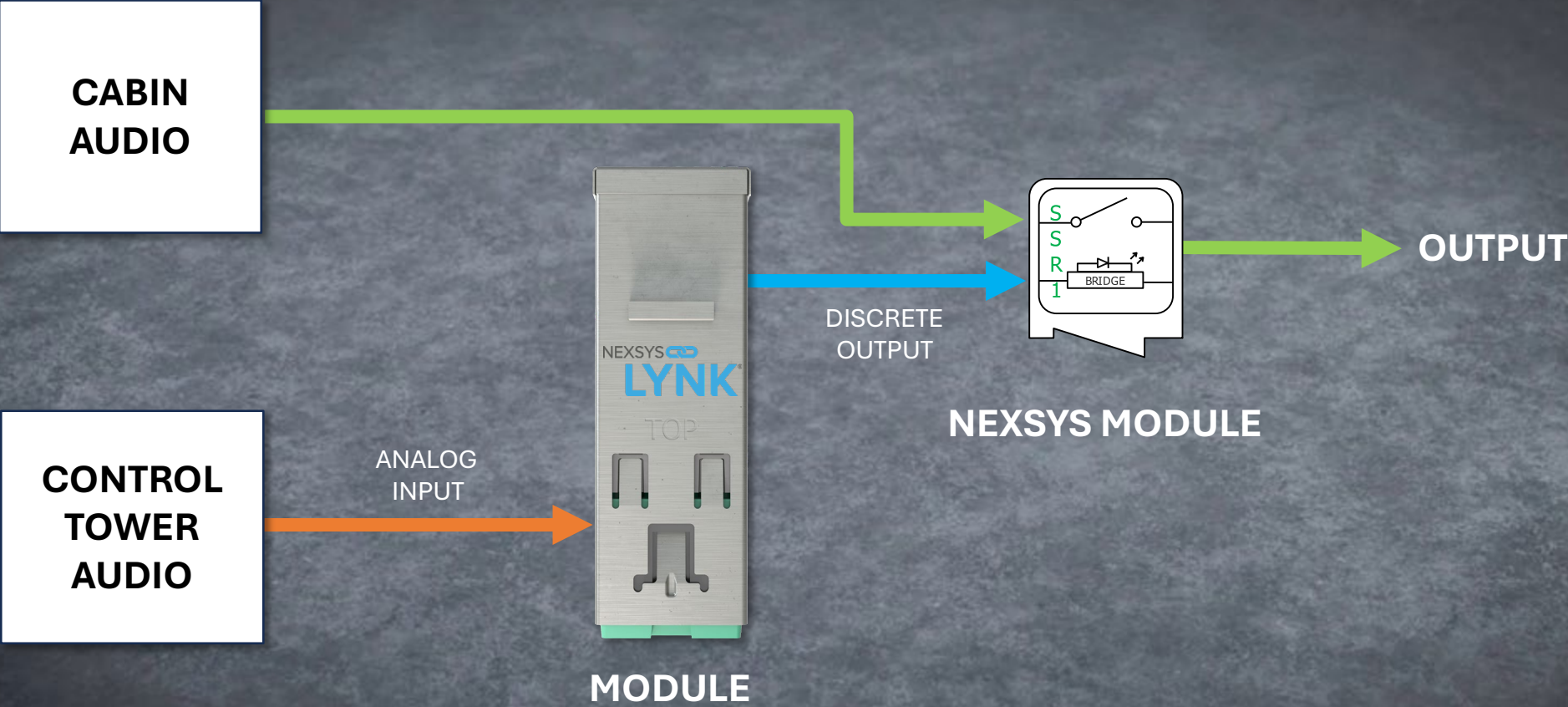
MFD Flight Phase-Based Aural Alert Controller

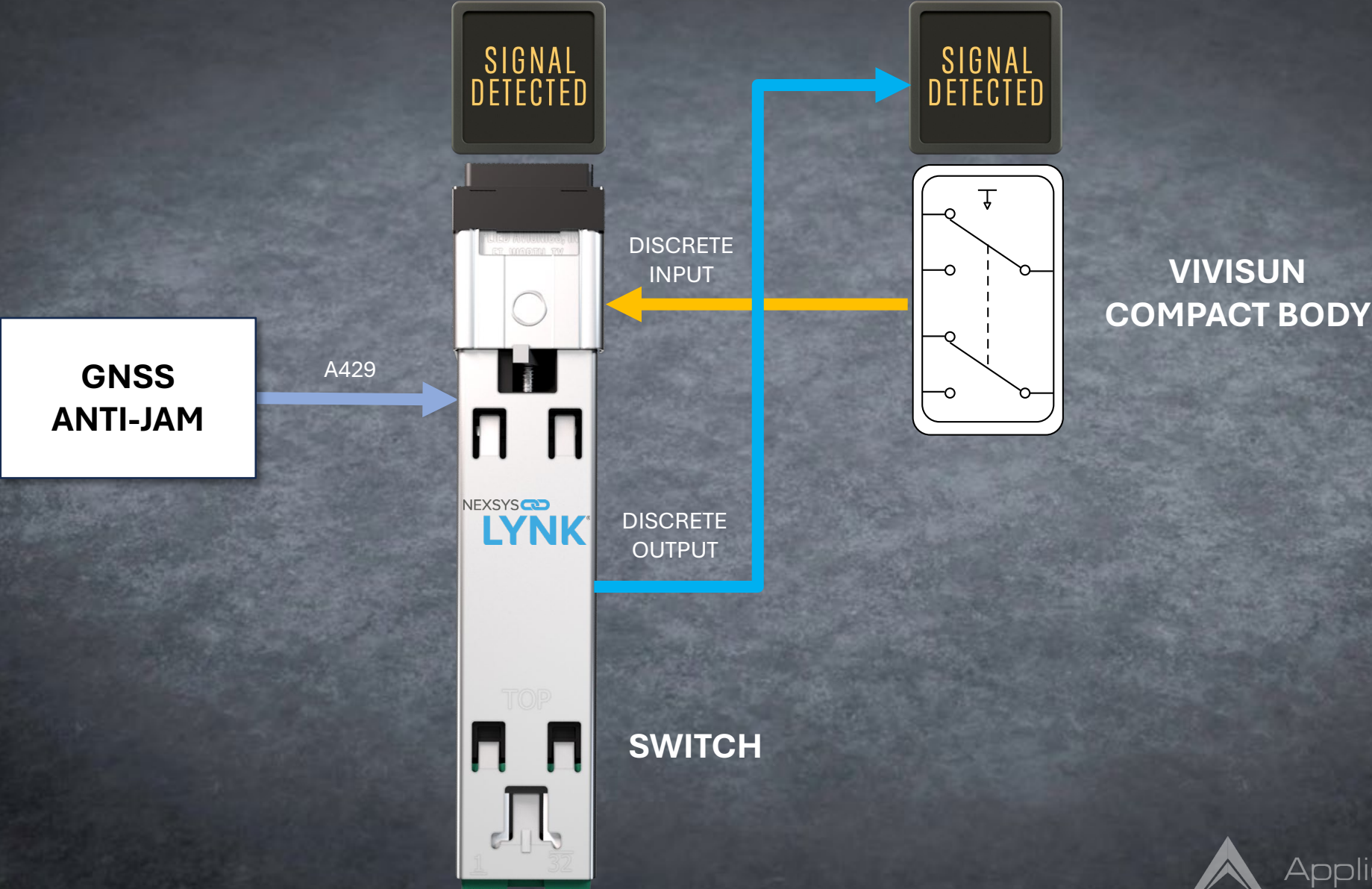




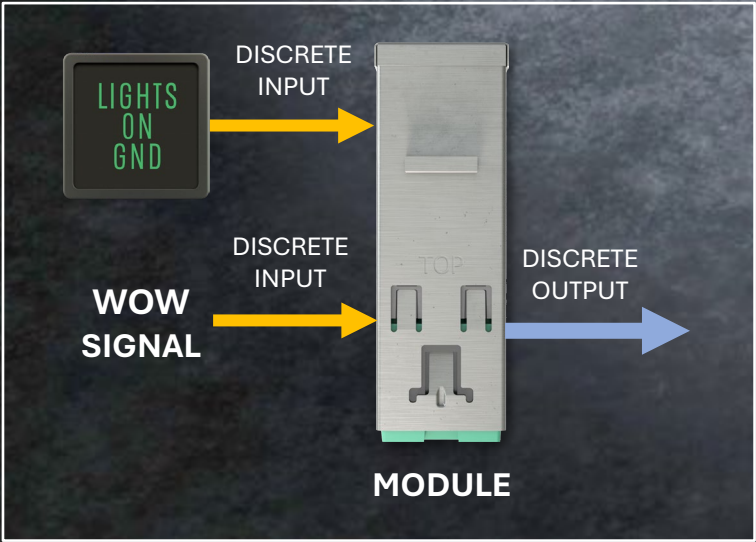
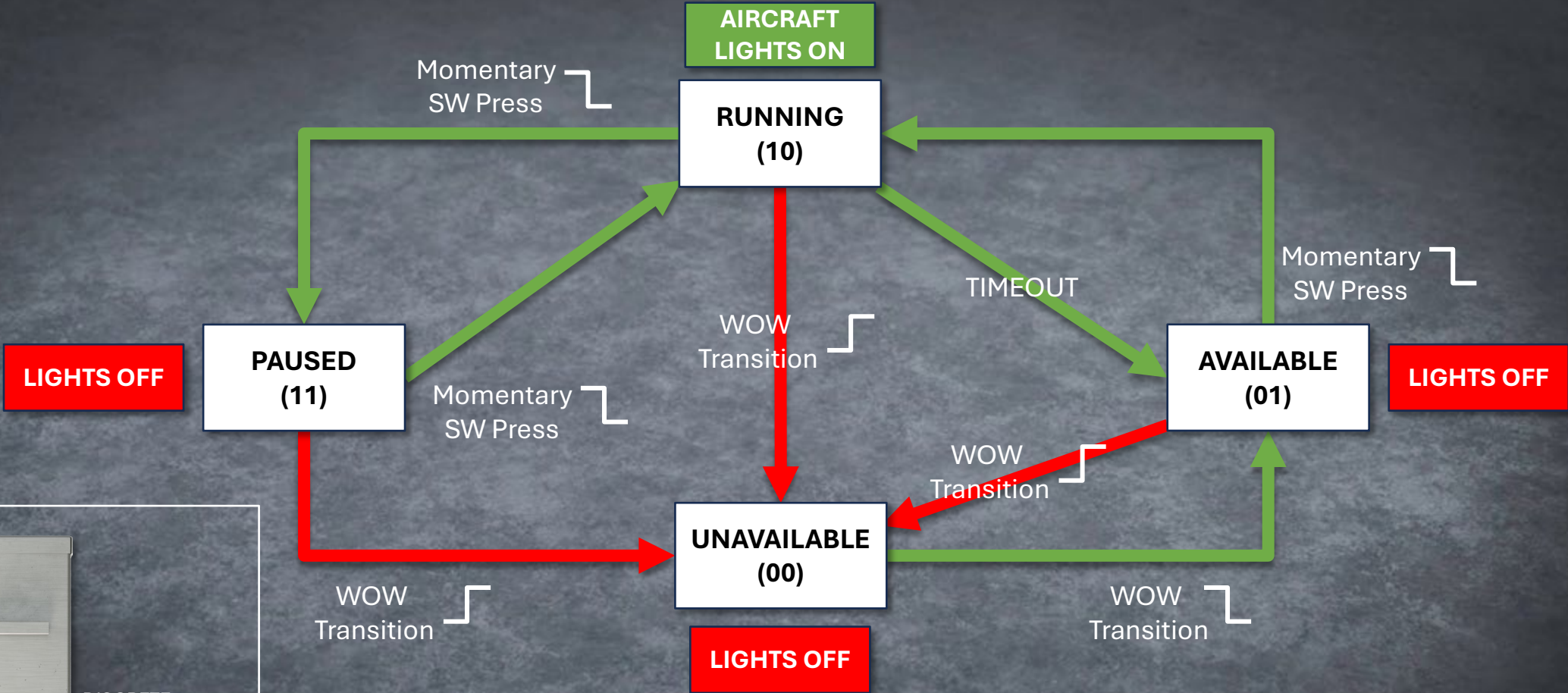


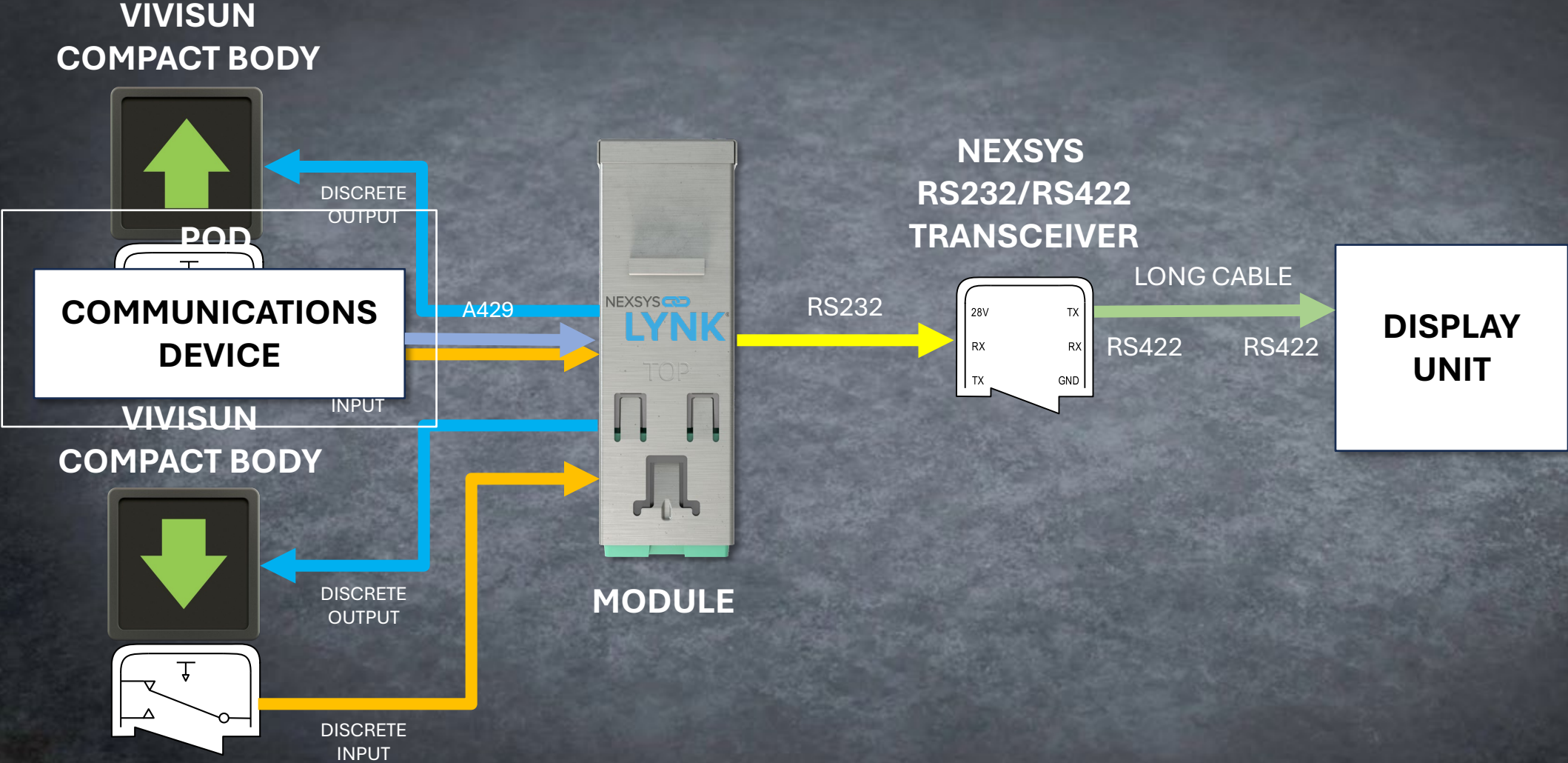
Analog Input:	
Wide Range	[−11V to +32V] [−60V to +60V]
Medium Range	[−10V to +10V] [−20V to +20V]
Narrow Range	[−5V to +5V]

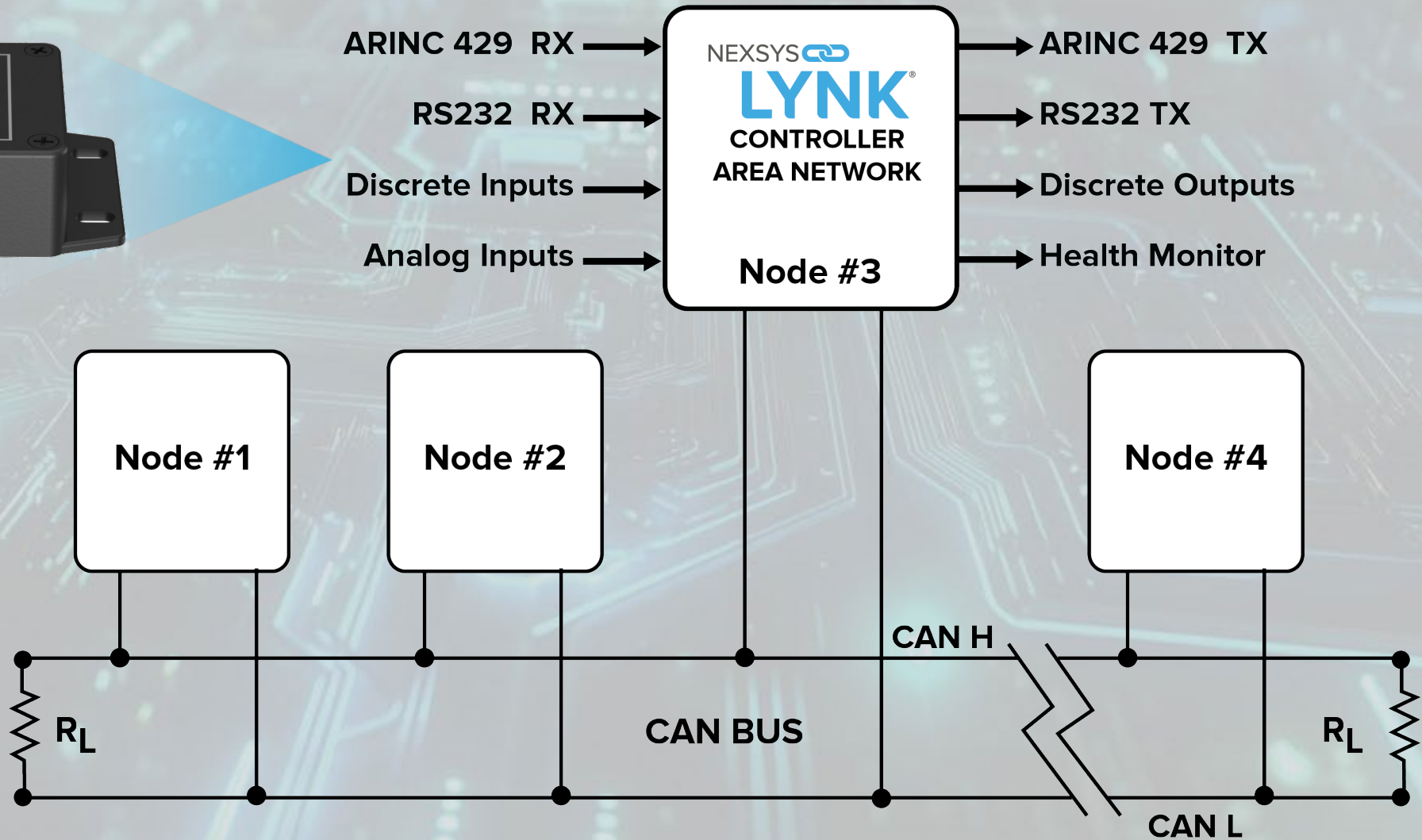


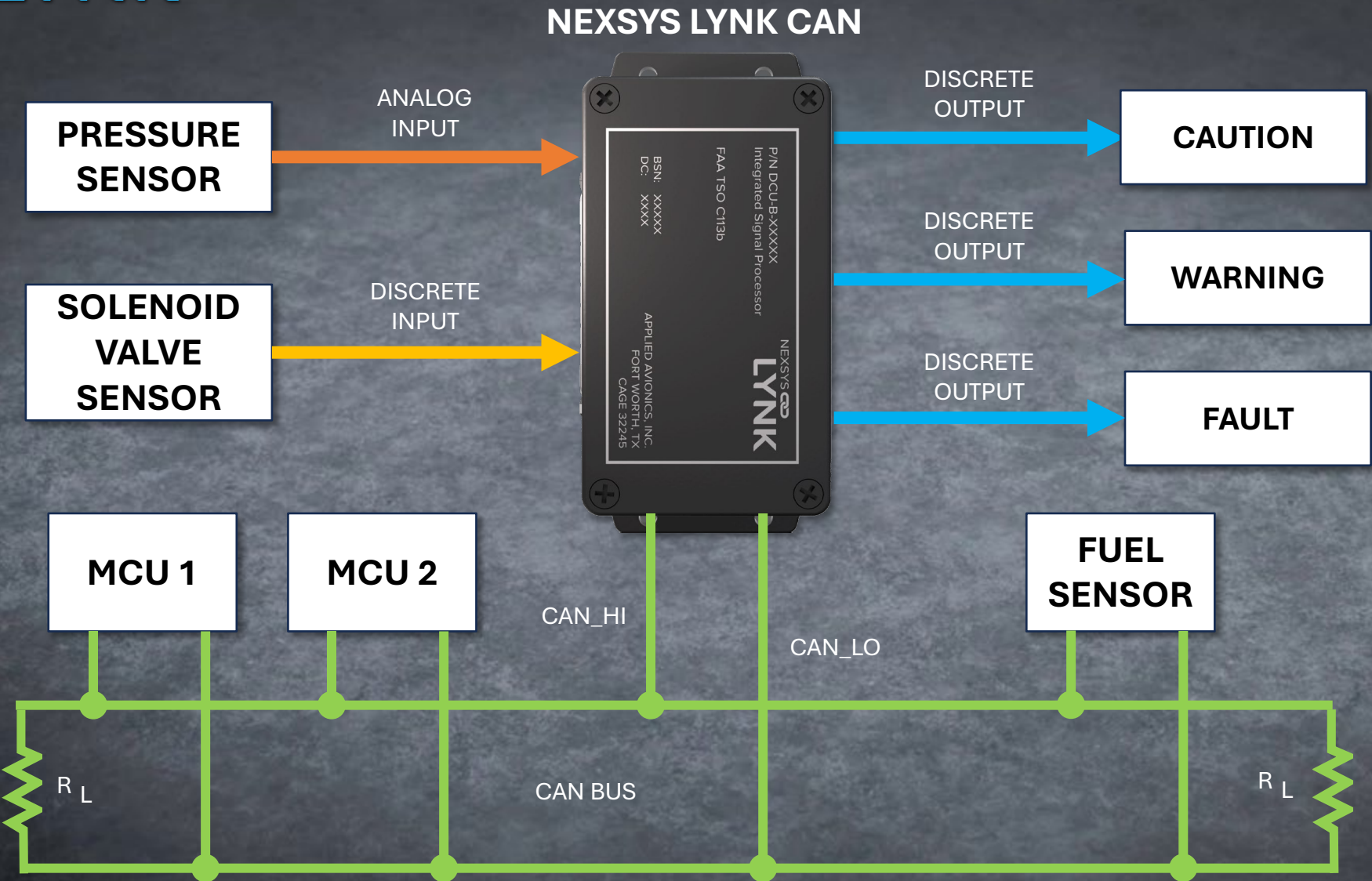


15 Minute Timer State Machine









ARINC429 RX/TX [High & Low Speed]

Discrete Input/Output

RS-232 RX/TX

CAN HI/LO

Pushbutton HMI

Health Monitor Output

Analog Input:

Audio Detection [-10V to +10V]

Wide Range [-11V to +32V]
[-60V to +60V]

Medium Range [-10V to +10V]
[-20V to +20V]

Narrow Range [-5V to +5V]

Upcoming [RS-485 RX/TX
RS-422 RX/TX



Solution Assessment

Establish Need & Alignment

An AAI Applications Engineer is assigned to partner with your team to understand objectives and align on the intended application



Requirements Gathering

Define Requirements & Confirm Solution Viability

AAI engineers collaborate closely with your team to fully define your unique Customer Application Specifications (CAS) and confirm the solution approach



Configuration Development

Create & Validate the Specific Solution

AAI engineers create the custom configuration, develop test scripts, and complete quality approvals



Evaluation Unit

Build, Test & Approve

AAI builds and tests an evaluation unit and delivers it to you for your verification and validation testing



Production Release

Transition to Full Production

The configuration is finalized, locked and fully certified production units are ready to ship



Applied Avionics

Thank you for attending.

Come See Us at Booth #806